

MOEC: Equipping Indonesian youths for the future of learning with big data

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MoECRT builds it Education Platform Compute Engine innovation in education services and analytics on BigQuery to improve student learning and policy making



About Ministry of Education, Culture, Research and Technology (MoECRT)

The Ministry of Education, Culture, Research and Technology (MoECRT) equips students with skills for global competitiveness, as well as values aligned with Indonesia's culture and ethics. In addition, the Ministry promotes the country's scientific, technological, and research capabilities to achieve economic progress. Taking on the role of chair of the G20 Culture

Google Cloud results

- Scales test delivery for hundreds of thousands of students on Compute Engine
- Flexible cloud billing accelerates digital innovation for government agency on a fixed annual budget
- Minimizes risk through compliance with Indonesia's public cloud requirements such as ISO/IEC 27001 and PP71

Enabling information education maker BigQuery

Ministers Meeting in September 2022, the Ministry will promote global collaboration to revitalize the arts and cultural sector, which has been severely affected by Covid-19.

Industries: Government & Public Sector

Location: Indonesia

Compute Engine



About Elitery

Elitery is a Google Cloud partner with years of experience managing mission-critical systems. The team has extensive experience working with clients across the public and private sectors. Elitery helps clients select the products and solutions they need to realize the best possible outcome in their digital transformation journey.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Elitery is a Google Cloud partner with years

Compute Engine

(<https://cloud.google.com/compute/>)

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Anyone who owns a mobile device can access information online, no matter where they are. With that in mind, Indonesia's Ministry of Education, Culture, Research and Technology (<https://pusdatin.kemdikbud.go.id/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

they need to realize the best possible

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

technology to accelerate the digitalization of education. Their goal is to help teachers and students reach their full potential by digitizing education processes.

With a super app in mind, MoECRT needed a scalable and reliable platform to serve more than three million teachers and 60 million students across Indonesia. To realize this ambition, MoECRT chose Google Cloud (<https://cloud.google.com/>) due to its performance capabilities and specifications.

"Google Cloud is a great platform that offers many features we need, and most importantly, it helps us comply with Indonesia's public cloud requirements such as ISO/IEC 27001 and PP71," says Aries Nugroho, IT Manager, ICT Center for Education at MoECRT. "With Google Cloud's local data centers, we can serve teachers and students from all over Indonesia with high availability."

MoECRT developed and launched its Digital Education Platform in August 2020. The education super app is the backbone for teachers, students, and schools to access digital services such as the National Assessment and National Talent Management. And having migrated to Google Cloud earlier, MoECRT already had the cloud foundation that enabled it to move quickly to digital learning

when schools in Indonesia closed earlier that year to contain the spread of COVID-19.

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Optimizing flexible resources for large-scale testing with Compute Engine

The National Assessment was previously a legacy workload hosted on an on-prem cloud to meet Indonesia government requirements. MoECRT had an annual budget, but did not have the flexibility needed to add servers and load balancers for the hundreds of thousands of students taking the national assessments.

"Cloud computing significantly reduces our fixed costs while improving our efficiencies since we no longer have to buy and manage physical servers," says Nugroho. "Along with Compute Engine (<https://cloud.google.com/compute>), Google Cloud offers useful data management products such as BigQuery (<https://cloud.google.com/bigquery>), which allows us to run queries on demand to make educational services more effective."

MoECRT completed a lift-and-shift migration to migrate 44 VMs from its on-premises server to Compute Engine (<https://cloud.google.com/compute>) in five short weeks, including testing. The revamped backend of the national assessment workload now scales up and down to handle traffic spikes when students from the 125,000 schools log in to take

their exams and afterward. MoECRT utilizes Cloud CDN (<https://cloud.google.com/cdn>) and Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to distribute content to students throughout Indonesia and handles 80,000 reads per second (RPS) at a peak without any bottlenecks.

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Closing learning gaps with data insights on BigQuery

Policymakers often rely on data to answer questions, such as how students do in different subject areas and how teacher competency correlates with student performance. This is why MoECRT collects data from students and teachers throughout their education journey, but it resides in multiple data silos. The data team now uses BigQuery to spot patterns and inform educational policy decisions by combining various data sources, such as student and teacher data.

Although strategic data, such as the teacher database, is stored on-premises, transactional data such as learner interactions with the platform content, is increasing in volume by the day, making cloud hosting a more cost-effective alternative.

"As we can run queries in BigQuery on demand, we don't need many database engineers to manage data," says Nugroho. "The heavy lifting is handled by Google Cloud, and we focus on the analysis to improve our education services."

MoECRT also uses BigQuery to measure school performance based on a Minimum Competency assessment. It uses test scores to gauge the depth of learning in each school, unlike national examinations that measure a student's

performance. With these insights, the ministry assigns mentor teachers with the proper skillset to disadvantaged schools and areas, by cross-referencing the teacher database.

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*—Aries Nugroho, IT
Manager, ICT Center for
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Tracking Bantuan Operasional Sekolah (BOS) funds distributed to schools for educational equipment such as laptops, projectors, and Wi-Fi routers is another use case for BigQuery. Different amounts are allocated depending on the number of students in each school. Previously, school administrators had to submit paper receipts for purchases made from this school operation assistance fund, often lost or misplaced. MoECRT uses BigQuery to analyze how BOS funds are impacting learning outcomes and increase transparency of education spending.

"From teachers to private enterprises, everyone in the community can contribute to our students' success with our super app," says Nugroho. "By transforming the way we deliver learning and measure educational outcomes, Google Cloud helps us continuously improve the quality of education."